

Covid-19 Model Prediction

Summary of Work:

In this project, we aimed to predict Covid-19 outcomes using a KNN classification model. The dataset included key features such as age, symptoms, medical history, and exposure history to predict the likelihood of infection and severity of symptoms. After preprocessing the data, including feature scaling and handling missing values, we split the data into 70% training and 30% evaluation sets. The KNN algorithm was used due to its simplicity and effectiveness in making predictions based on similarity to neighboring data points. The model achieved an accuracy of 70%, which can be improved through various future enhancements like optimizing the value of K, handling imbalanced data, and applying dimensionality reduction techniques.

Technologies/Libraries and Specific Tasks:

1. **Python** – Used as the primary programming language for building the KNN model and processing the dataset.
2. **Pandas** – Utilized for data manipulation and preprocessing, including handling missing values, encoding categorical variables, and splitting the dataset.
3. **NumPy** – Used for numerical operations, including array manipulation and performing mathematical computations on the dataset.
4. **Matplotlib/Seaborn** – Employed for data visualization, including plotting histograms, scatter plots, and heatmaps to understand the data and the model's performance.

Target Audience:

- Data scientists and machine learning engineers working on healthcare prediction models.
- Researchers and analysts interested in applying machine learning to epidemiology and public health data.
- Developers and data engineers involved in building predictive models for Covid-19 or similar health-related outcomes.